

# A Comparative Study of Convolutional and Transformer Architectures Under Adversarial Perturbations: Gradient Characteristics and Transferability Analysis

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**Abstract**—How convolutional neural networks (CNNs) and Vision Transformers (ViTs) differ under adversarial perturbations remains an open question. This paper presents a behavioral comparison of the two architecture families under matched standard adversarial training, pairing robustness scores with gradient- and feature-level analyses. Using CIFAR-10 with adversarially trained ResNet-18 and ViT-Tiny models evaluated under AutoAttack at the standard perturbation budget of  $8/255$  on the full test set, we report four findings. First, the CNN attains significantly higher robust accuracy than the ViT (35.74 percent versus 32.94 percent; McNemar paired test on 10,000 samples,  $p$  below 0.001), a modest but reliable 2.8-point gap. Second, cross-architecture transfer between this model pair is symmetric once each target’s clean error is controlled for (conditioned fooling rates of 20.95 versus 20.32 percent, statistically equivalent within a two-point margin), whereas the widely used raw transfer metric manufactures an apparent 8.3-point asymmetry—a methodological caution for transfer studies comparing targets of unequal clean accuracy. Third, scale-invariant sparsity measures show CNN gradients concentrate sensitivity more locally (Hoyer sparsity 0.474 versus 0.449), a difference that persists in a native-resolution ViT control and is therefore not an input-upsampling artifact, while cross-sample gradient alignment is low for both models but higher for the ViT (0.052 versus 0.038). Fourth, ViT feature representations under attack exhibit an early-drop, mid-network-plateau degradation profile, with clean-adversarial cosine similarity falling from 0.994 to 0.955 by mid-network while feature norms change by less than 1.5 percent. These characterizations inform the evaluation and deployment of mixed-architecture ensembles.

**Index Terms**—Adversarial robustness, convolutional neural networks, deep learning security, transfer attacks, Vision Transformer

## I. INTRODUCTION

Deep neural networks have achieved remarkable success across various computer vision tasks, including image classification, object detection, and semantic segmentation. However, these models are vulnerable to adversarial examples—carefully crafted perturbations that are imperceptible to humans but can cause misclassification with high confidence [1], [2]. This vulnerability poses significant security concerns for safety-critical applications such as autonomous driving, medical diagnosis, and security systems.

Two dominant architectural paradigms have emerged in computer vision: Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs). CNNs leverage local connectivity and weight sharing through convolutional operations, capturing hierarchical spatial features effectively [3]. In contrast, ViTs apply the self-attention mechanism [4] to image patches, enabling global context modeling from early layers [5]. While both architectures achieve competitive performance on clean data, their behavior under adversarial attacks exhibits fundamental differences that are not yet fully understood.

Previous research has extensively studied adversarial robustness for CNNs, leading to effective defense mechanisms such as adversarial training [6] and TRADES [7]. However, the comparative analysis of CNN and ViT robustness remains limited, with existing studies often focusing on *whether* one architecture is more robust than another, rather than *why* they exhibit different behaviors. Understanding the mechanistic factors underlying these differences is crucial for developing more secure deep learning systems.

In this work, we move beyond simple robustness comparisons to characterize how CNNs and ViTs respond differently to adversarial perturbations. Our investigation centers on several interconnected questions: How do adversarial perturbations transfer between architectures, and does any apparent asymmetry survive once differences in the targets’ clean accuracy are controlled for? Do gradient characteristics correlate with the observed robustness differences? How do adversarial perturbations propagate through Vision Transformer layers, and what feature-level dynamics lead to misclassification? What practical guidance for evaluating and deploying mixed-architecture ensembles follows from these findings?

This paper makes several contributions to understanding architectural differences in adversarial vulnerability. We provide a comprehensive robustness evaluation using AutoAttack [8], the gold standard benchmark, comparing adversarially trained ResNet-18 and ViT-Tiny under  $\epsilon = 8/255$   $L_\infty$  perturbations. Our transfer attack analysis characterizes cross-architecture transferability using a clean-accuracy-controlled (conditioned) metric, showing that raw transfer rates can be misleading when the compared targets differ in clean accuracy: an apparent 8-point transfer asymmetry between our models vanishes entirely under the conditioned metric. We find that CNN gradients concentrate sensitivity more locally under scale-invariant spar-

sity measures—a difference that survives a native-resolution ViT control—while cross-sample gradient alignment is low for both architectures but moderately higher for the ViT. Furthermore, complementing earlier layer-wise robustness studies of ImageNet-scale ViTs [9], [10], we present a systematic layer-wise analysis of how adversarial perturbations affect the feature representations of an *adversarially trained* ViT, revealing an early-drop, mid-network-plateau degradation profile in which most directional deviation accumulates by mid-network rather than growing monotonically with depth. To support reproducible research, our complete experimental pipeline, trained models, and analysis tools will be made publicly available upon acceptance (see Data and Code Availability).

We begin by reviewing related work on adversarial attacks, defenses, and architecture-specific robustness studies in Section II. Section III describes our experimental methodology, and Section IV presents results across four complementary analyses: robustness comparison, transfer attacks, gradient characteristics, and feature degradation. We discuss the implications of our findings in Section V and conclude in Section VI.

## II. RELATED WORK

### A. Adversarial Attacks

Adversarial attacks aim to find minimal perturbations that cause misclassification. The Fast Gradient Sign Method (FGSM) [2] generates perturbations in a single step by following the sign of the loss gradient, with the perturbation magnitude bounded under the  $L_\infty$  norm. Projected Gradient Descent (PGD) [6] extends FGSM through iterative optimization, taking multiple gradient-sign steps and projecting back onto the  $\epsilon$ -ball after each step (formal definitions in Section III-C).

The Carlini & Wagner (C&W) attack [11] formulates adversarial example generation as an optimization problem, often achieving higher success rates but at increased computational cost. More recently, AutoAttack [8] combines multiple attack strategies (APGD-CE, APGD-T, FAB-T, Square Attack) to provide a reliable robustness evaluation, becoming the de facto standard for benchmarking.

### B. Defense Mechanisms

Adversarial training (AT) [6] remains the most effective defense, training models on adversarial examples generated during each batch:

$$\min_{\theta} \mathbb{E}_{(x,y) \sim \mathcal{D}} \left[ \max_{\|\delta\|_\infty \leq \epsilon} \mathcal{L}(f_{\theta}(x + \delta), y) \right] \quad (1)$$

TRADES [7] provides a theoretically-grounded trade-off between natural accuracy and robustness:

$$\mathcal{L}_{\text{TRADES}} = \mathcal{L}_{CE}(f(x), y) + \beta \cdot \text{KL}(f(x) \| f(x_{adv})) \quad (2)$$

where  $\beta$  controls the trade-off parameter.

MART [12] addresses the issue of misclassified examples by emphasizing them during training. Other approaches include input transformation [13], certified defenses via randomized smoothing [14], and ensemble methods [15]. Recent advances

in certified robustness include Adaptive Randomized Smoothing [16], which improves certification radii through multi-step defenses.

### C. CNN Robustness

Extensive research has characterized the adversarial robustness of CNN architectures. Madry et al. [6] demonstrated that adversarial training with PGD attacks provides strong empirical robustness. The RobustBench leaderboard [17] tracks state-of-the-art CNN robustness, with WideResNet variants achieving the highest robust accuracy on CIFAR-10.

Studies have shown that model capacity correlates with achievable robustness [6], with wider and deeper networks demonstrating improved robust accuracy. Architecture-specific findings include the role of skip connections in adversarial transferability [18] and the distinct batch-normalization statistics that adversarial training induces at scale [19]; relatedly, auxiliary batch-normalization branches allow adversarial examples to improve clean accuracy [20]. Adversarial training is also known to reshape input-gradient structure itself, yielding more perceptually aligned [21] and sparser [22] gradients.

### D. Vision Transformer Robustness

Vision Transformers [5], building on the self-attention mechanism introduced by Vaswani et al. [4], process images as sequences of patches; their internal representations differ measurably from CNNs' layer by layer [23]. Initial studies suggested that ViTs exhibit some inherent robustness properties due to their global receptive field [9]. Bai et al. [24] systematically investigated whether transformers are more robust than CNNs, finding that under fair training recipes transformers are not more adversarially robust than CNNs, while exhibiting superior robustness under distribution shift attributed to self-attention; Benz et al. [25] reported an early direct comparison including MLP-Mixer and linked the differences to frequency preferences. Our work extends this line of inquiry by analyzing adversarially trained models of comparable scale and providing behavioral characterizations through gradient characteristics and feature degradation analysis. Subsequent work revealed that ViTs remain vulnerable to adversarial attacks, particularly gradient-based methods [26], with patch-targeted attacks exploiting the attention mechanism directly [27] and attention shown to shift measurably under patch perturbations [28].

Shao et al. [10] demonstrated that adversarial training can improve ViT robustness—and that ViT features rely less on high-frequency content—though challenges remain due to training instability. ViT robustness to common corruptions and occlusions has also been characterized [29]. More recently, Jain et al. [30] proposed Adaptive Attention Scaling Adversarial Training (AAS-AT) to address ViT-specific training challenges, achieving state-of-the-art robustness on CIFAR-10 and ImageNet-100; with diffusion-augmented adversarial training, ViTs have even been reported to surpass WideResNets on small-scale datasets [31]. Zhu et al. [32] strengthened transferable attacks on ViTs through gradient normalization scaling and high-frequency adaptation. Despite these advances,

comprehensive behavioral comparisons between CNN and ViT robustness under unified evaluation frameworks remain limited.

### E. Transfer Attacks

Adversarial examples often transfer between models, enabling black-box attacks [33]. The transferability depends on factors including model architecture, training data, and attack methods [34]. Foundational transfer studies already evaluate transferability on correctly classified samples [35], and standard attack papers condition their success rates for the same reason [36]; recent evaluation guidelines make such protocol choices explicit [37]. In ensemble design, Ravikumar et al. [38] restrict transfer measurement to samples correctly classified by both models precisely to account for accuracy differences between them—the same confound our conditioned metric controls.

Cross-architecture transfer between CNNs and ViTs has received increasing attention. Mahmood et al. [26] measured cross-architecture transfer with a both-correct protocol and found notably low CNN $\leftrightarrow$ ViT transferability; Naseer et al. [39] showed that adversarial examples crafted on ViTs transfer poorly under conventional attacks and proposed self-ensemble and token refinement to improve ViT-sourced transferability, and Wei et al. [40] improved ViT-sourced transfer by bypassing attention gradients and patch-wise dropout. Building on these, Token Gradient Regularization (TGR) [41] crafts transferable adversarial samples by exploiting ViT’s attention components, while Momentum Integrated Gradients (MIG) [42] achieves improved transferability across both ViTs and CNNs. Chen et al. [43] systematically analyze the factors affecting cross-architecture transferability, finding that architectural inductive biases—CNNs preferring high-frequency information while ViTs favor low-frequency patterns—significantly impact transfer success. Understanding these transfer patterns is crucial for developing robust ensemble systems and evaluating real-world attack scenarios.

### F. Research Gap

While individual studies have examined CNN and ViT robustness separately, and the pitfalls of unfair CNN–transformer comparisons have been highlighted before [24], [44], existing comparative work typically reports robustness scores in isolation—often with inconsistent perturbation budgets, attack configurations, or model scales—rather than pairing the comparison with behavioral analysis. A recent study in this journal compares ViT and CNN adversarial robustness directly [45]; our work differs by evaluating on the full test set with AutoAttack and paired statistics, by controlling transfer measurement for clean-accuracy differences, and by accompanying the comparison with gradient- and feature-level behavioral analyses.

Our work also engages the measurement question directly. Conditioning transfer evaluation on correctly classified samples is established practice [26], [35], [36], [38]; our contribution is to quantify, in a controlled same-experiment comparison, how large a spurious cross-architecture asymmetry the

unconditioned raw metric can manufacture (8.3 percentage points in our setting) when targets differ in clean accuracy. We investigate gradient characteristics alongside transfer behavior, analyze layer-wise feature degradation in ViTs under adversarial perturbations, and connect these findings to practical guidance for evaluating and deploying mixed-architecture ensembles.

## III. METHODOLOGY

### A. Threat Model

We consider the standard  $L_\infty$  threat model where an adversary can perturb input images within a bounded region:

$$\mathcal{B}_\epsilon(x) = \{x' : \|x' - x\|_\infty \leq \epsilon\} \quad (3)$$

Following established conventions for CIFAR-10 [17], [46], we use  $\epsilon = 8/255 \approx 0.031$  as the primary perturbation budget, with additional full-test PGD-10 evaluations at  $\epsilon \in \{2/255, 4/255, 16/255\}$  for the epsilon sweep analysis.

We evaluate under both white-box (full model access) and black-box (transfer attack) scenarios. For white-box attacks, the adversary has complete knowledge of model architecture and parameters. For transfer attacks, adversarial examples are generated on a source model and evaluated on a different target model.

### B. Model Architectures

a) *CNN Models*: **ResNet-18** [3]: A widely-used baseline CNN with residual connections, containing 11.2M parameters. We use the standard architecture designed for CIFAR-10 (initial  $3 \times 3$  convolution instead of  $7 \times 7$ ).

**WideResNet-28-10** [47]: A wider variant of ResNet with 36.5M parameters, representing the state-of-the-art architecture for adversarial robustness on CIFAR-10. We use the pretrained robust checkpoint of Goyal et al. [48] distributed via RobustBench [17]; note that this model was trained with additional unlabeled data and a stronger recipe than our standard adversarial training, so it serves as a reference point rather than a matched comparison.

b) *Vision Transformer Models*: **ViT-Tiny**: We use the ViT-Tiny architecture from the `timm` library [49], adapted for CIFAR-10 by bilinearly upsampling inputs to  $224 \times 224$ . The architecture uses  $16 \times 16$  patches (196 patches per image), 192-dimensional embeddings, 12 transformer blocks with 3 attention heads each, MLP ratio of 4, and totals 5.7M parameters. The model is trained *from scratch* on CIFAR-10 (no ImageNet pretraining) with basic augmentation (random crop and horizontal flip), then adversarially fine-tuned. Since ViT robustness on small datasets is known to be sensitive to the training recipe [50], [51], our architecture-level conclusions should be interpreted as holding *under this baseline training recipe* rather than for optimally trained ViTs.

### C. Attack Methods

a) *FGSM*: Single-step attack with step size equal to the perturbation budget:

$$x_{adv} = x + \epsilon \cdot \text{sign}(\nabla_x \mathcal{L}_{CE}(f(x), y)) \quad (4)$$

b) *PGD*: Iterative attack with step size  $\alpha = 2/255$  and random initialization:

$$x^{t+1} = \Pi_{\mathcal{B}_\epsilon(x)}(x^t + \alpha \cdot \text{sign}(\nabla_{x^t} \mathcal{L}_{CE}(f(x^t), y))) \quad (5)$$

with  $x^0 = x + \mathcal{U}(-\epsilon, \epsilon)$ . We use PGD-10 (10 iterations) for both adversarial training and evaluation throughout this work (Table I).

c) *AutoAttack*: We use AutoAttack [8], the standardized ensemble attack combining APGD-CE (Auto-PGD with cross-entropy loss), APGD-T (Auto-PGD with targeted DLR loss), FAB-T (targeted Fast Adaptive Boundary attack), and Square Attack (score-based black-box attack). This ensemble represents the gold standard for robustness evaluation.

#### D. Defense Methods

a) *Adversarial Training (AT)*: We employ a mixed adversarial training objective that combines clean and adversarial losses:

$$\mathcal{L} = (1 - \lambda) \mathcal{L}_{CE}(f_\theta(x), y) + \lambda \mathcal{L}_{CE}(f_\theta(x_{adv}), y) \quad (6)$$

where  $\lambda = 0.5$  balances standard classification performance with adversarial robustness. The inner maximization generates adversarial examples using PGD-10 ( $\alpha = 2/255$ ,  $\epsilon = 8/255$ ), computed with the model in training mode so that attack generation and the parameter update see consistent batch-normalization statistics, following standard adversarial-training implementations [52]. This mixed objective, widely adopted in practice, helps maintain clean accuracy while building robustness, as training solely on adversarial examples can degrade natural performance. While alternative formulations such as TRADES [7] offer theoretically grounded trade-off mechanisms, this work focuses on standard adversarial training to isolate architectural effects from defense-specific interactions.

#### E. Training Protocol

a) *Clean Training*: For CNNs, we use SGD optimizer with initial learning rate 0.1, cosine annealing schedule [53], weight decay  $5 \times 10^{-4}$ , and 200 epochs. For ViTs, we use AdamW optimizer [54] with initial learning rate  $10^{-3}$ , cosine annealing, weight decay 0.05, and 200 epochs.

b) *Adversarial Training*: Starting from our clean-trained checkpoints, we fine-tune with reduced learning rates: CNNs use SGD with LR  $10^{-3}$  and cosine annealing for up to 100 epochs, while ViTs use AdamW with the same learning rate and schedule. Gradients are clipped to a global  $L_2$  norm of 10, and batches with non-finite loss are skipped. Data augmentation includes random cropping with 4-pixel padding and horizontal flipping; all models consume raw pixel inputs in  $[0, 1]$  without mean/std normalization, and attacks and epsilon budgets operate directly in this pixel space. The training batch size is 128 for the CNN and 64 for the ViT (evaluation batch sizes vary by analysis). Model selection and early stopping use PGD-10 adversarial accuracy on a held-out 2,000-sample validation split carved from the training set, so the test set is touched only by the final post-hoc evaluations. The training seeds are 42 for ResNet-18 and 123 for ViT-Tiny; since

the validation permutation is derived from the training seed, the two models use different (fixed) held-out splits. Because the validation images were seen during clean pretraining, validation accuracies are optimistic in absolute terms, but this does not affect test-set results. Early stopping uses patience 20 with a 0.1 percentage-point improvement threshold: the ResNet-18 run stopped at epoch 84 and the ViT-Tiny run at epoch 51, truncating their cosine schedules.

#### F. Evaluation Metrics

a) *Robustness Metrics*: We evaluate models using three standard metrics: clean accuracy (classification performance on unperturbed test images), robust accuracy (classification performance on adversarial examples), and attack success rate (fraction of correctly classified clean images that become misclassified after attack).

b) *Transfer Attack Metrics*: Our primary transfer metric is the *conditioned fooling rate*: among the samples that the target model classifies correctly on clean inputs, the fraction that becomes misclassified under adversarial examples crafted on the source model. Given adversarial examples  $\{x_{adv}^S\}$  generated against source model  $f_S$ , the conditioned transfer rate to target model  $f_T$  is:

$$T_c(f_S \rightarrow f_T) = \frac{\sum_{i=1}^n \mathbf{1}[f_T(x_i) = y_i] \mathbf{1}[f_T(x_{adv,i}^S) \neq y_i]}{\sum_{i=1}^n \mathbf{1}[f_T(x_i) = y_i]} \quad (7)$$

Conditioning transfer evaluation on correctly classified samples follows established practice [26], [35]–[38] and is essential in cross-architecture comparisons: the raw misclassification rate  $\frac{1}{n} \sum_i \mathbf{1}[f_T(x_{adv,i}^S) \neq y_i]$  includes the target’s clean error, so a target with lower clean accuracy appears more “vulnerable” to transfer even when no additional samples are flipped by the attack. We report the raw rate as a secondary metric for transparency and to quantify the size of this confound. Transfer rates are computed on the full test set (10,000 samples) with per-sample outcomes logged, and we report 95% confidence intervals obtained by percentile bootstrap over samples (1,000 resamples).

c) *Gradient Analysis Metrics*: We analyze gradient properties through three complementary metric families, computing gradients of per-sample losses so that magnitudes are independent of the evaluation batch size. Gradient norm ( $\|\nabla_x \mathcal{L}\|_2$  and  $\|\nabla_x \mathcal{L}\|_\infty$ ) measures overall sensitivity magnitude. Gradient sparsity is quantified with scale-invariant measures—Hoyer sparsity, the Gini coefficient of  $|g|$ , and the fraction of components below 1% of the per-sample maximum—since fixed absolute thresholds conflate gradient scale with structure; Hoyer and Gini satisfy the scale-invariance axioms formalized by Hurley and Rickard [55]. Gradient alignment measures the average pairwise absolute cosine similarity across samples (computed over all within-batch pairs):

$$\text{Alignment} = \frac{1}{N(N-1)} \sum_{i \neq j} \left| \frac{\nabla_{x_i} \mathcal{L} \cdot \nabla_{x_j} \mathcal{L}}{\|\nabla_{x_i} \mathcal{L}\| \|\nabla_{x_j} \mathcal{L}\|} \right| \quad (8)$$

High alignment indicates similar perturbation directions across inputs, suggesting vulnerability to universal adversarial perturbations.

d) *Feature Degradation Metrics*: To analyze how adversarial perturbations affect ViT representations across layers, we compute three metrics at each layer  $\ell$ . Feature cosine similarity measures semantic preservation:

$$\text{Sim}_\ell = \frac{h_\ell(x) \cdot h_\ell(x_{adv})}{\|h_\ell(x)\| \|h_\ell(x_{adv})\|} \quad (9)$$

where  $h_\ell$  denotes the representation at layer  $\ell$ . Feature norm change captures magnitude effects:

$$\Delta_\ell = \frac{\|h_\ell(x_{adv})\| - \|h_\ell(x)\|}{\|h_\ell(x)\|} \times 100\% \quad (10)$$

Feature  $L_2$  distance ( $\|h_\ell(x) - h_\ell(x_{adv})\|_2$ ) quantifies absolute deviation.

### G. Statistical Validation

To assess evaluation stochasticity, we evaluate each trained model 3 times with different random seeds (42, 123, 456), reporting mean  $\pm$  standard deviation. The evaluated subset and its ordering are fixed (the test loader does not shuffle), so the seeds vary *only* the random initialization of PGD restarts; the measured variance therefore quantifies attack-initialization stochasticity, not data-sampling or training variance. To address training-level reproducibility, we conducted multiple independent adversarial-training runs per architecture: two under a preliminary implementation of the training loop, and one final run per architecture under the standardized protocol reported in this paper—rankings were preserved across all runs, with non-negligible run-to-run variation for the ViT (Section IV-E). All evaluation and analysis scripts fix all random seeds (seed 42, including CUDA determinism flags), so every reported number is exactly reproducible from the released pipeline; the 3-seed analysis above bounds the residual attack-initialization stochasticity that an unseeded rerun would exhibit at roughly  $\pm 0.1$  percentage points. All experiments are conducted on NVIDIA RTX 5060 Ti (16GB) using PyTorch 2.6.0 with CUDA 12.8.

## IV. EXPERIMENTS

### A. Robustness Comparison

Table I presents the main robustness comparison across all evaluated models. We report clean accuracy and robust accuracy under PGD-10 and AutoAttack with  $\epsilon = 8/255$ .

The results indicate a statistically significant robustness advantage for the CNN under standard adversarial training: on the full test set, ResNet-18 AT achieves 35.74% robust accuracy under AutoAttack, compared to 32.94% for ViT-Tiny AT—a 2.80 percentage-point gap. Because both models are evaluated on the same 10,000 samples with per-sample outcomes logged, we can apply McNemar’s paired test: 940 samples are robustly classified only by the CNN versus 660 only by the ViT ( $\chi^2 = 48.65$ ,  $p < 10^{-11}$ ), so the gap is not attributable to evaluation noise. The 95% binomial confidence half-widths are below  $\pm 1$  percentage point per model. Two caveats remain. First, the gap is modest relative to training-level variability: across our training runs the ViT’s PGD-10 accuracy varied by roughly four points (Section IV-E), so the

TABLE I: Main Robustness Results on CIFAR-10 ( $\epsilon = 8/255$ ). All cells for our models are measured post hoc on the full 10,000-sample test set with fully seeded evaluations (seed 42, deterministic CUDA). Checkpoints were selected on a held-out validation split (Section III-E). AA = AutoAttack (standard suite).

| Model                            | Defense         | Params | Clean | FGSM  | PGD-10 | AA           |
|----------------------------------|-----------------|--------|-------|-------|--------|--------------|
| <i>CNN Models</i>                |                 |        |       |       |        |              |
| ResNet-18                        | None            | 11.2M  | 94.37 | 30.93 | 0.01   | –            |
| ResNet-18                        | AT              | 11.2M  | 85.42 | 49.90 | 40.97  | <b>35.74</b> |
| WRN-28-10                        | AT <sup>†</sup> | 36.5M  | 89.48 | 70.91 | 66.92  | 62.76        |
| <i>Vision Transformer Models</i> |                 |        |       |       |        |              |
| ViT-Tiny                         | None            | 5.7M   | 77.50 | 3.34  | 0.03   | –            |
| ViT-Tiny                         | AT              | 5.7M   | 75.65 | 40.07 | 36.09  | <b>32.94</b> |

<sup>†</sup>Robust checkpoint of Goyal et al. [48] via RobustBench [17].

Clean/FGSM/PGD-10: our local full-test evaluations; AA: the RobustBench-reported figure (we did not rerun AutoAttack on it).

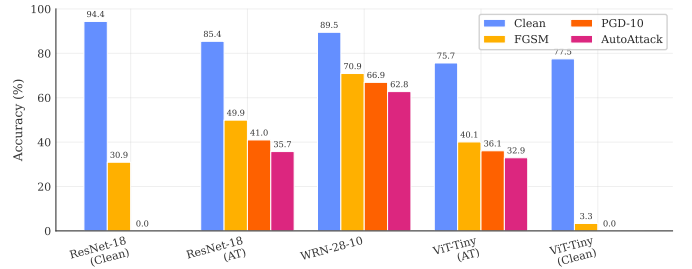


Fig. 1: Clean, FGSM, PGD-10, and AutoAttack accuracy at  $\epsilon = 8/255$  on the full CIFAR-10 test set. CNNs outperform ViTs under standard adversarial training; WideResNet-28-10 achieves the highest robust accuracy through capacity, additional training data, and a stronger recipe jointly [48] (its AutoAttack bar is the RobustBench-reported value).

architecture comparison should be read as “under matched training recipes” rather than as a universal constant. Second, the CNN also retains a higher clean accuracy (85.42% vs 75.65%, a 9.8-point gap), reflecting the difficulty of training ViTs on small datasets from scratch. AutoAttack robustness is consistently lower than PGD-10 robustness for both architectures (5.23 points for ResNet-18, 3.15 points for ViT-Tiny), confirming the importance of using this stronger benchmark. The WideResNet-28-10 baseline [48], at 62.76% robust accuracy, demonstrates how far capacity, additional training data, and a stronger training recipe jointly push robustness beyond architectural choice alone.

Figure 1 visualizes this robustness gap across architectures. To further understand the effect of perturbation magnitude, Figure 2 shows PGD-10 robust accuracy across perturbation budgets  $\epsilon \in \{2/255, 4/255, 8/255, 16/255\}$  on the full test set.

Figure 3 provides visual examples of adversarial perturbations and their effects on model predictions.

### B. Transfer Attack Analysis

We analyze the transferability of adversarial examples between architectures using PGD-10 attacks on the full test

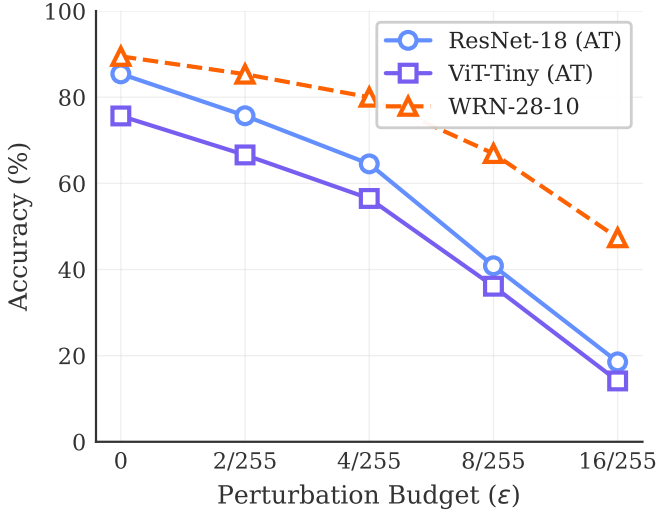


Fig. 2: PGD-10 robust accuracy vs. perturbation budget ( $\epsilon$ ), measured on the full test set. Both AT architectures degrade with similar shapes as epsilon increases, with the CNN maintaining a consistent margin.



Fig. 3: Adversarial example visualization. Each row shows: original image with model prediction, perturbation magnified  $10\times$ , and adversarial image with new prediction. Row 1: Attack on ResNet-18. Row 2: Attack on ViT-Tiny. Row 3: Transfer attack (generated on ResNet, evaluated on ViT). The perturbations are imperceptible yet cause confident misclassifications.

set ( $n=10,000$ ). Table II reports the conditioned fooling rate defined in Section III-F—the fraction of samples the target classifies correctly on clean inputs that become misclassified under adversarial examples crafted on the source—with raw misclassification rates in parentheses for transparency.

TABLE II: Transfer Attack Matrix on the full test set ( $n=10,000$ ). Cells show the conditioned fooling rate with 95% bootstrap confidence intervals (10,000 resamples); raw target-misclassification rates in parentheses. Diagonal entries are white-box attacks.

| Source $\rightarrow$ Target | ResNet-18 AT             | ViT-Tiny AT              |
|-----------------------------|--------------------------|--------------------------|
| ResNet-18 AT                | 52.2 [51.1, 53.3] (59.1) | 21.0 [20.0, 21.9] (40.2) |
| ViT-Tiny AT                 | 20.3 [19.5, 21.2] (31.9) | 52.3 [51.3, 53.4] (63.9) |

The conditioned analysis yields two findings. First, once the targets’ unequal clean accuracy is controlled for, cross-architecture transfer between this model pair (under PGD-10 at  $\epsilon = 8/255$ ) is *symmetric*: CNN-to-ViT fooling is 20.95% [20.03, 21.88] and ViT-to-CNN fooling is 20.32% [19.47, 21.19], a 0.63-point difference that is not significant under a two-proportion test ( $z = 0.98$ ,  $p = 0.33$ ) and, more stringently, is formally *equivalent* within a  $\pm 2$ -point margin under a two one-sided test (TOST,  $p = 0.016$ ). The symmetry is not an artifact of the attack choice: repeating the analysis with the momentum-based MI-FGSM [36] (10 steps,  $\mu=1.0$ ) yields conditioned rates of 19.85% [18.97, 20.78] and 20.11% [19.28, 20.96], again with fully overlapping intervals. Nor is it a triviality of the metric: applying the identical protocol to the *clean-trained* counterparts yields much lower conditioned transfer (9.91% [9.32, 10.53] CNN $\rightarrow$ ViT vs 11.76% [11.15, 12.41] ViT $\rightarrow$ CNN)—a small but statistically distinguishable asymmetry, while the raw metric would report a spurious 12.2-point gap in the *opposite* direction (28.3% vs 16.1%). Matched adversarial training thus both increases cross-architecture transferability and eliminates the residual directional difference present between the clean models. Second, the raw rates paint a misleadingly different picture (40.2% vs 31.9%, an apparent 8.3-point asymmetry favoring CNN sources): because the ViT target has roughly ten points more clean error than the CNN target (75.65% vs 85.42% clean accuracy), the raw metric credits the CNN source with failures the ViT would have made on clean inputs anyway. This is a cautionary methodological result for cross-architecture transfer studies: apparent transfer asymmetries between targets of unequal clean accuracy can be manufactured entirely by the metric. Notably, the white-box conditioned fooling rates are almost identical across architectures (52.2% vs 52.3%), indicating that—under matched adversarial training—the two architectures are comparably vulnerable on the samples they classify correctly.

Figure 4 visualizes the conditioned transfer matrix.

### C. Gradient Characteristics

To characterize architectural differences in input sensitivity, we analyze the gradient properties of both architectures, together with a CIFAR-native ViT control ( $32\times 32$  inputs,  $4\times 4$  patches, no upsampling) trained with the same adversarial protocol, which isolates the effect of the  $32\rightarrow 224$  upsampling in our main ViT pipeline. Table III summarizes the key statistics using scale-invariant sparsity measures (Section III-F);

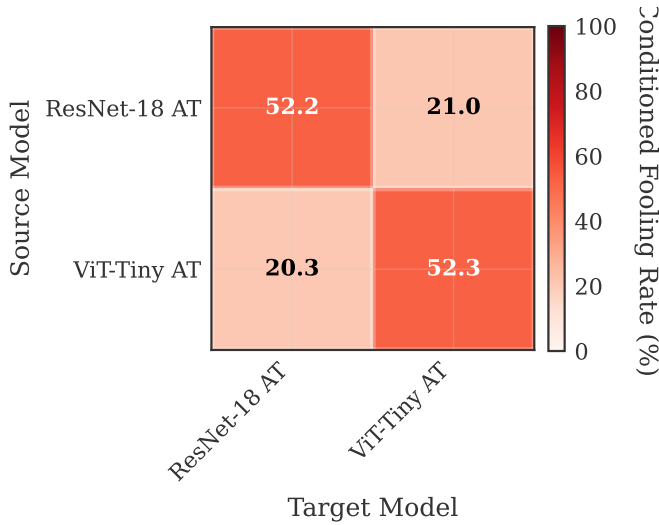


Fig. 4: Conditioned transfer fooling-rate heatmap ( $n=10,000$ ). Off-diagonal transfer is symmetric once target clean error is controlled for (21.0% vs 20.3%).

TABLE III: Gradient Characteristics ( $n=500$  test samples; per-sample loss gradients; alignment is the all-pairs mean absolute cosine similarity within batches of 50;  $\pm$  values are std across 10 batch means). All ResNet–ViT differences on the scale-invariant metrics are significant at the batch level (Welch  $t$ -test,  $p < 3 \times 10^{-5}$ ).

| Metric                 | ResNet-18 AT             | ViT-Tiny AT              | ViT (native) <sup>‡</sup> |
|------------------------|--------------------------|--------------------------|---------------------------|
| $L_2$ Norm (mean)      | 0.698                    | 0.760                    | 0.554                     |
| $L_\infty$ Norm (mean) | 0.139                    | 0.144                    | 0.087                     |
| Hoyer Sparsity         | <b>0.474</b> $\pm$ 0.009 | 0.449 $\pm$ 0.011        | 0.405 $\pm$ 0.008         |
| Gini Coefficient       | <b>0.634</b> $\pm$ 0.006 | 0.606 $\pm$ 0.008        | 0.571 $\pm$ 0.006         |
| Frac. <1% of max       | <b>31.5%</b> $\pm$ 1.7   | 25.8% $\pm$ 1.7          | 17.9% $\pm$ 1.2           |
| Gradient Alignment     | 0.038 $\pm$ 0.002        | <b>0.052</b> $\pm$ 0.002 | 0.079 $\pm$ 0.005         |

<sup>‡</sup>CIFAR-native ViT control ( $4 \times 4$  patches, no upsampling; 5.4M params; checkpoint-selection clean/robust acc. 62.4%/31.8%). Included only to test the upsampling confound.

gradients are computed with per-sample losses so that norms do not depend on the evaluation batch size.

The gradient analysis reveals consistent but modest architectural differences. Across all three scale-invariant sparsity measures, CNN gradients are sparser than ViT gradients (Hoyer 0.474 vs 0.449; Gini 0.634 vs 0.606; 31.5% vs 25.8% of components below 1% of the per-sample maximum), indicating that CNN sensitivity is somewhat more concentrated in localized regions. Interestingly, this ordering is a product of adversarial training: in the clean-trained counterparts the ViT is the sparser model (Hoyer 0.406 vs 0.354), and adversarial training sparsifies the CNN far more strongly (+0.12 Hoyer) than the ViT (+0.04), consistent with the known sparsifying effect of adversarial training on input gradients [22]. The native-resolution ViT control is *less* sparse than the upsampled ViT on every measure, so the CNN–ViT sparsity ordering is not an artifact of the  $32 \rightarrow 224$  upsampling—if anything, upsampling moves the ViT *toward* the CNN. We emphasize that these differences are far smaller than a threshold-based sparsity metric would suggest: an earlier version of this

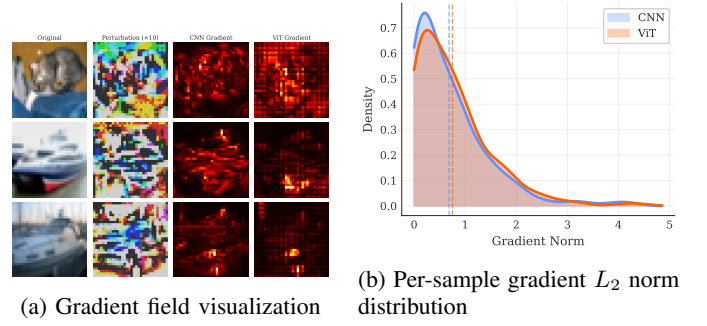


Fig. 5: Gradient characteristics comparison. (a) Input-gradient magnitude maps for representative test images with the corresponding PGD-10 perturbations. (b) Distribution of per-sample gradient  $L_2$  norms for both architectures.

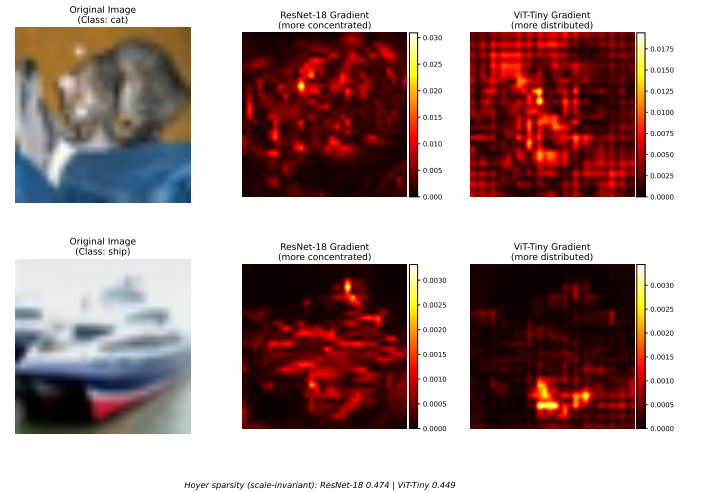


Fig. 6: Gradient heatmap comparison between CNN and ViT on the same input images. CNN gradients (top) are modestly more concentrated around object edges, while ViT gradients (bottom) are somewhat more spatially distributed (Hoyer sparsity annotations from the gradient-analysis artifacts).

analysis (fixed  $10^{-6}$  threshold, measured on the preliminary run2 models) indicated a  $1.7\times$  sparsity ratio, most of which reflected gradient-scale differences rather than structure; on the final models the same scale-dependent metric inflates the ratio even further ( $\approx 2.4\times$ ). Cross-sample gradient alignment is low for both architectures but measurably higher for the ViT (0.052 vs 0.038, a  $1.36\times$  ratio; native control 0.079), suggesting that transformer sensitivity directions are somewhat more shared across inputs. We treat both characteristics as behavioral correlates of, rather than established explanations for, the architectures’ differing adversarial behavior.

Figure 5 provides visual evidence for these gradient differences.

Figure 6 shows a direct comparison of gradient heatmaps, highlighting the structural differences in how each architecture computes input gradients.

TABLE IV: ViT-Tiny Feature Degradation Under PGD-10 Attack ( $n=100$  test samples; features are MLP sub-block outputs;  $\pm$  values are std across 5 batch means)

| Layer    | $L_2$ Distance  | Cosine Sim.                         | Norm Change        |
|----------|-----------------|-------------------------------------|--------------------|
| Block 0  | $2.73 \pm 0.13$ | $0.994 \pm 0.000$                   | $-0.11 \pm 0.17\%$ |
| Block 1  | $1.74 \pm 0.07$ | $0.992 \pm 0.001$                   | $-0.05 \pm 0.22\%$ |
| Block 2  | $1.81 \pm 0.05$ | $0.981 \pm 0.002$                   | $-0.99 \pm 0.31\%$ |
| Block 3  | $1.33 \pm 0.05$ | $0.974 \pm 0.002$                   | $-0.69 \pm 1.24\%$ |
| Block 4  | $1.71 \pm 0.08$ | $0.972 \pm 0.002$                   | $-1.07 \pm 1.76\%$ |
| Block 5  | $1.92 \pm 0.03$ | $0.972 \pm 0.003$                   | $-0.29 \pm 1.54\%$ |
| Block 6  | $1.84 \pm 0.04$ | $0.961 \pm 0.004$                   | $-0.28 \pm 0.83\%$ |
| Block 7  | $1.48 \pm 0.06$ | $0.963 \pm 0.004$                   | $-1.40 \pm 0.99\%$ |
| Block 8  | $1.79 \pm 0.03$ | <b><math>0.955 \pm 0.003</math></b> | $-1.05 \pm 0.27\%$ |
| Block 9  | $1.86 \pm 0.02$ | $0.963 \pm 0.002$                   | $-0.73 \pm 0.54\%$ |
| Block 10 | $1.73 \pm 0.09$ | $0.965 \pm 0.003$                   | $-0.76 \pm 0.45\%$ |
| Block 11 | $2.31 \pm 0.15$ | $0.965 \pm 0.004$                   | $-1.11 \pm 0.80\%$ |

#### D. Feature Degradation in Vision Transformers

We analyze how adversarial perturbations affect ViT feature representations across layers, extending layer-wise representation analyses of clean-trained ViTs [23] to the adversarially trained setting. The measured representations are the outputs of each transformer block’s MLP sub-block; the attention-weight analysis accompanies the figure-level results. Table IV shows all twelve blocks ( $n=100$  test samples).

The layer-wise analysis reveals a characteristic degradation profile. Cosine similarity between clean and adversarial features drops quickly in the early blocks (0.994 at Block 0 to 0.972 by Block 4), reaches its minimum in the middle-to-late network (0.955 at Block 8), and then plateaus with a slight recovery in the final blocks (0.965 at Blocks 10–11). Most of the directional deviation is therefore accumulated by mid-network rather than growing monotonically to the output—an “early-drop, mid-network-plateau” profile rather than strictly progressive degradation. The final-block recovery is statistically reliable at the batch level (Block 8 vs Block 11 cosine similarity, Welch  $t = -4.07$ ,  $p = 0.005$ ). A parallel analysis of the adversarially trained ResNet-18’s eight residual-block outputs (same attack, same metrics,  $n=100$ ) shows a qualitatively different profile: cosine similarity degrades *monotonically* with depth (0.995 at layer1.0 down to 0.913 at layer4.1, with no plateau or recovery), accompanied by much larger norm distortions in the final blocks ( $-6.4\%$  at layer4.0,  $+7.2\%$  at layer4.1). The plateau-and-recovery shape with preserved norms is therefore specific to the ViT in our setting, not a generic property of deep networks under attack. Feature norm changes remain modest throughout (at most  $-1.4\%$ ), indicating that adversarial perturbations corrupt the *direction* of representations far more than their magnitude. The mild recovery in the final blocks may reflect the regularizing influence of the classifier head on the last transformer blocks. Even so, the terminal deviation (cosine similarity  $\approx 0.965$ ) is sufficient to flip predictions on most attacked samples: on these 100 samples, accuracy drops from 79% clean to 39% under PGD-10.

Complementing the feature-level measurements, Figure 7 visualizes CLS-token attention maps at layers 1, 6, and 12 for a representative sample: early-layer attention is nearly unchanged under attack, while later layers show visibly larger

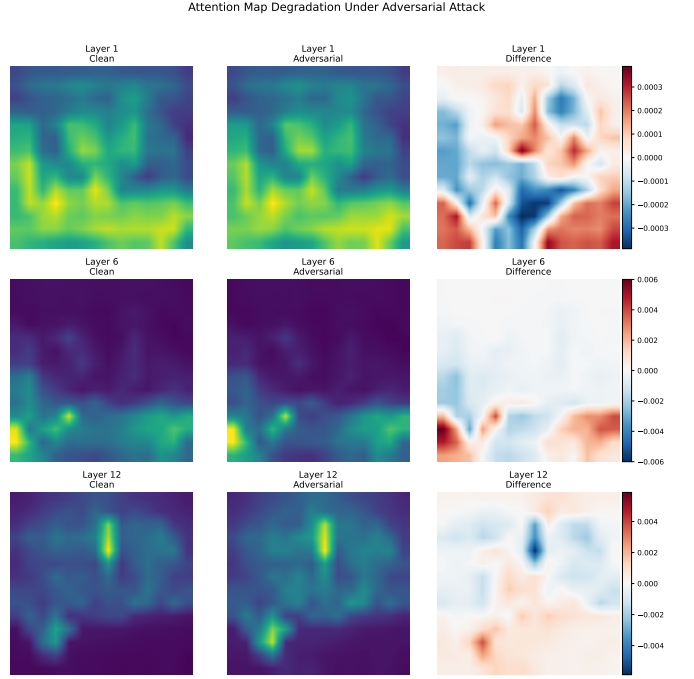


Fig. 7: CLS-token attention maps (post-softmax, head-averaged) across ViT layers under adversarial attack. Each row shows a different layer (1, 6, 12); columns show clean attention, adversarial attention, and their difference. Early layers maintain similar attention patterns, while later layers show larger attention shifts.

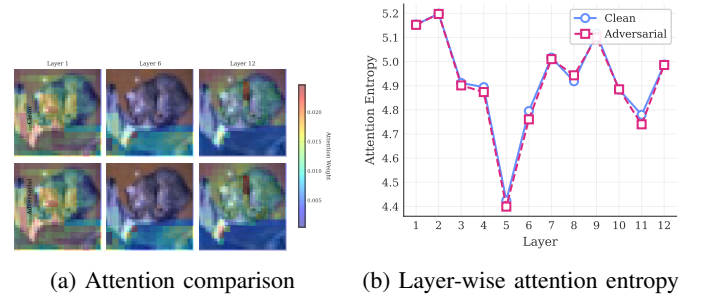


Fig. 8: ViT attention analysis under attack: (a) clean vs. adversarial CLS attention for a representative sample; (b) per-layer entropy of the CLS attention distribution for clean and adversarial inputs.

attention shifts, consistent with attention-shift observations under patch perturbations [28] and with attacks that explicitly target the attention mechanism [27].

Figure 8 complements the feature-level analysis with attention-level measurements. Notably, the per-layer *entropy* of the CLS attention distribution is nearly unchanged under attack (mean absolute change  $\approx 0.015$  nats on layer means of  $\approx 4.9$  nats, at most 0.8%; measured on the 8-sample figure batch): the attack redirects *where* attention goes (Figure 7) without broadening or sharpening its distribution, reinforcing that the operative degradation is directional drift in the feature space rather than attention diffusion.

Figure 9 provides a t-SNE visualization of how adversarial

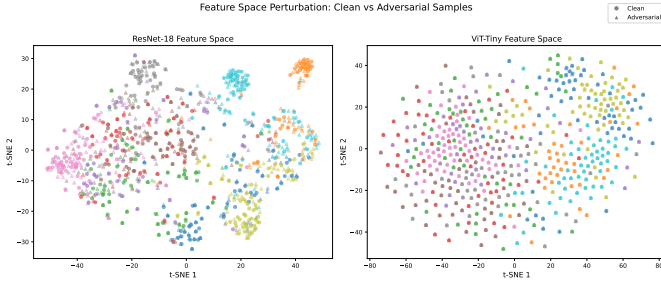


Fig. 9: t-SNE visualization of feature space perturbation. Clean samples (circles) vs adversarial samples (triangles) for ResNet-18 (left) and ViT-Tiny (right). Adversarial perturbations cause feature displacement that differs between architectures: CNN features show more class-coherent displacement, while ViT features exhibit greater inter-class confusion.

TABLE V: Statistical Validation (Mean  $\pm$  Std over 3 evaluation seeds, first 1,000 test samples)

| Model        | Clean Acc (%)    | PGD-10 Acc (%)   |
|--------------|------------------|------------------|
| ResNet-18 AT | 86.10 $\pm$ 0.00 | 40.57 $\pm$ 0.12 |
| ViT-Tiny AT  | 76.90 $\pm$ 0.00 | 38.23 $\pm$ 0.09 |

perturbations affect the learned feature space. The visualization reveals that adversarial examples shift features away from their original class clusters, with different displacement patterns for each architecture.

#### E. Statistical Validation

To quantify evaluation stochasticity, we evaluate each trained model across 3 different random seeds (42, 123, 456) for attack initialization on 1,000 test samples. Table V shows the results.

The low variance confirms that evaluation stochasticity is negligible. The evaluated subset and its ordering are identical across seeds (the test loader does not shuffle), so clean accuracy shows exactly zero variance and the only stochastic element is the random initialization of PGD restarts. Robust accuracy shows minimal variation (standard deviation 0.12 percentage points for CNN, 0.09 for ViT), indicating stable attack behavior across random restarts. This procedure quantifies attack-initialization stochasticity only; it does not measure data-sampling or training-level variance. (Differences between Table V and Table I PGD values—0.4 points for the CNN and up to 2.1 points for the ViT—reflect the easier first-1,000-sample subset versus the full test set.)

Beyond evaluation seeds, we conducted multiple independent adversarial-training runs per architecture during this study. Under the final training protocol, ResNet-18 AT reached 40.97% and ViT-Tiny AT 36.09% full-test PGD-10 accuracy; earlier runs under a preliminary implementation of the training loop (differing in attack-generation details, later standardized as described in Section III-D) produced values in the 40.3–41.0% (CNN) and 32.8–36.9% (ViT) ranges. The CNN-over-ViT ranking was preserved in every run, though the ViT’s run-to-run variation is clearly larger than evaluation-level variance,

and we therefore avoid attributing precise magnitudes to the architecture gap beyond the paired analysis of Section IV-A.

**Computational Details:** All experiments were conducted on an NVIDIA RTX 5060 Ti (16GB VRAM) using PyTorch 2.6.0 with CUDA 12.8. AutoAttack evaluation used the standard configuration (APGD-CE, APGD-T, FAB-T, Square) on the full 10,000-sample test set, evaluated in seeded 1,000-sample chunks with per-sample outcome logging; total wall-clock time was several hours per model on this hardware.

## V. DISCUSSION

### A. Key Findings and Their Implications

*a) Cross-Architecture Transfer Is Symmetric Once Clean Accuracy Is Controlled:* Our experiments quantify a methodological point that we believe deserves more attention in cross-architecture transfer studies. Conditioning transfer evaluation on correctly classified samples is established practice [35], [36], [38]; what our controlled same-experiment comparison adds is a measurement of how much the unconditioned raw metric can distort the picture. Under the raw metric, our models exhibit a substantial apparent transfer asymmetry (40.2% CNN $\rightarrow$ ViT vs 31.9% ViT $\rightarrow$ CNN on the full test set). Under the conditioned fooling-rate metric, the asymmetry vanishes: 20.95% [20.03, 21.88] vs 20.32% [19.47, 21.19], a difference that is formally equivalent within a  $\pm 2$ -point margin (TOST,  $p = 0.016$ ). The raw gap is manufactured by the ten-point clean-accuracy difference between the targets, which the raw metric credits to the attacking source. We therefore report cross-architecture transfer between this adversarially trained model pair—under PGD-10 at  $\epsilon = 8/255$ —as *symmetric*, and we caution that raw-rate comparisons between targets of unequal clean accuracy can produce spurious directional conclusions. Interestingly, the conditioned white-box rates are also nearly identical across architectures (52.2% vs 52.3%), reinforcing that on commonly solved samples the two architectures are comparably attackable under matched training.

This null result carries interpretive weight: since conditioned transfer is symmetric in our setting, the gradient-level differences we measure (sparsity, alignment) evidently do not translate into directional transfer advantages at this scale. We note that genuine architectural transfer asymmetries can exist—Mahmood et al. [26] observe asymmetries even under a both-correct protocol—so our result does not imply that all reported asymmetries are metric artifacts; rather, it shows the clean-accuracy confound must be excluded before an asymmetry is interpreted architecturally.

*b) Gradient Characteristics as Vulnerability Signatures:* The gradient analysis characterizes how the two architectures distribute input sensitivity. Across three scale-invariant measures [55], CNN gradients are consistently sparser than ViT gradients (Hoyer 0.474 vs 0.449; Gini 0.634 vs 0.606; 31.5% vs 25.8% of components below 1% of the per-sample maximum), indicating modestly more localized sensitivity—consistent with the known tendency of adversarial training to sparsify input gradients [22] and to reshape gradient structure more broadly [21], here differing in degree between architectures. Two controls strengthen this reading. First, the native-resolution ViT control is even less sparse than the upsampled

ViT, so the ordering is not an artifact of input upsampling. Second, the effect size is honest: an earlier threshold-based analysis (on the preliminary run2 models) suggested a  $1.7\times$  sparsity ratio, most of which we traced to gradient-scale differences rather than structure—the structural difference is real but small. In parallel, cross-sample gradient alignment is measurably higher for the ViT (0.052 vs 0.038, all-pairs absolute cosine), suggesting somewhat more shared sensitivity directions across inputs, though both values are low in absolute terms.

The alignment result merits careful interpretation. Universal adversarial perturbations [56] exploit sensitivity directions shared across inputs; one might therefore hypothesize that ViTs’ global self-attention produces higher cross-sample gradient alignment. We observe a difference in that direction—ViT alignment is  $1.36\times$  the CNN’s (0.052 vs 0.038), and the native-resolution ViT control is higher still (0.079)—but the absolute levels are low for both architectures, far from the strong alignment that would make universal perturbations cheap. Measuring the same quantity in the clean-trained counterparts resolves the training-regime question directly: clean alignment is 0.050 for the ViT and 0.030 for the CNN, essentially the same levels as after adversarial training. The ViT–CNN alignment gap is therefore an *architectural* property that adversarial training neither creates nor suppresses appreciably in our setting—in contrast to gradient magnitude, which adversarial training reduces by an order of magnitude for both models, and to sparsity, whose architecture ordering adversarial training reverses (Section IV-C).

*c) Feature Degradation in Vision Transformers:* Our layer-wise analysis reveals a characteristic feature degradation profile in ViT layers. In the first two transformer blocks, adversarial inputs produce noticeable  $L_2$  distance from clean representations while maintaining semantic direction (cosine similarity above 0.99) and near-constant norms; the perturbation initially displaces representations without appreciably altering either their norm or their semantic direction. Directional deviation then accumulates rapidly through the early-to-middle network, with cosine similarity reaching its minimum of 0.955 at Block 8, after which it plateaus with a mild recovery in the final blocks (0.965 at Blocks 10–11), plausibly reflecting the regularizing influence of the classifier head. The degradation is subtler than the attack’s effectiveness might suggest—norm changes remain below 1.5% throughout—implying that modest directional shifts, largely established by mid-network, are sufficient to flip predictions on most attacked samples.

This early-drop, mid-network-plateau profile suggests that effective defenses should focus on stabilizing early-to-middle layer representations rather than only filtering perturbations at the input level or protecting the final layers. Intermediate supervision or layer-wise consistency regularization might help maintain semantic integrity throughout the network. The parallel residual-block analysis of the adversarially trained CNN (Section IV-D) confirms the profile is ViT-specific: CNN representations degrade monotonically with depth and with large terminal norm distortions, suggesting the two architectures would benefit from differently placed stabilization.

## B. Broader Implications

*a) For Researchers:* Our work demonstrates the value of mechanistic analysis over pure benchmarking. Simply reporting that one architecture achieves higher robust accuracy than another provides limited insight for improving defenses. Understanding *why* architectures behave differently—through gradient analysis, transfer studies, and feature degradation tracking—enables targeted improvements rather than trial-and-error defense development.

The consistent gap between AutoAttack and PGD-10 results (3.1–5.2 percentage points lower robust accuracy under AutoAttack for our models; Section IV-A) reinforces the importance of using standardized, strong evaluation protocols. While PGD-10 remains useful during development for its computational efficiency, final robustness claims should always be validated with AutoAttack or equivalent comprehensive evaluation. Beyond this, conditioned transfer rates measured from diverse source architectures could serve as a diagnostic for architecture-specific vulnerabilities—a direction our two-model study cannot itself establish, but one that the clean-accuracy-controlled methodology makes feasible.

*b) For Practitioners:* For deployment scenarios requiring adversarial robustness, our findings offer concrete guidance. CNNs currently provide better robust accuracy with more stable adversarial training dynamics, making them the safer choice when robustness is paramount. ViTs require more careful hyperparameter tuning during adversarial training and may benefit from specialized techniques beyond standard adversarial training.

When deploying ensemble defenses that combine CNN and ViT models, practitioners should evaluate transfer with clean-accuracy-controlled metrics before assuming either architecture is the harder surrogate; our analysis shows the apparent asymmetry arises entirely from clean-accuracy differences, and conditioned transfer between our models is symmetric. Additionally, the gap between WideResNet-28-10 (62.76% robust accuracy) and ResNet-18 (35.74%) shows how much headroom lies beyond our baselines—though this particular model owes its robustness to capacity, additional (unlabeled) training data, and a stronger training recipe jointly [48], so the gap should not be read as a capacity effect alone. For high-stakes applications, investing in larger models trained with strong recipes and augmented data may prove more effective than developing novel defense mechanisms.

*c) On Model Capacity:* Our comparison involves models with different parameter counts: ResNet-18 has 11.2M parameters while ViT-Tiny has 5.7M. This  $2\times$  capacity difference could partially explain the robustness gap. However, our mechanistic analyses suggest that architectural factors play a more fundamental role than model size alone.

While parameter count differs (11.2M vs. 5.7M), our gradient analysis reveals scale-invariant structural differences—modestly sparser CNN gradients and moderately higher ViT alignment—that are consistent with architectural processing differences and survive the native-resolution control. At the same time, the symmetric conditioned transfer result shows these gradient-level differences do not produce directional transfer effects, so architectural attribution of the

robust-accuracy gap itself remains partial: capacity, clean-accuracy differences, and training-recipe maturity for ViTs on small datasets all plausibly contribute. We acknowledge that capacity-controlled experiments (e.g., using ViT-Small with  $\sim 22\text{M}$  parameters) would provide stronger evidence for disentangling these factors. We leave such experiments for future work.

### C. Limitations

Several limitations constrain the scope of our conclusions and suggest directions for future validation.

Our ViT-Tiny model uses  $224 \times 224$  bilinear upsampling from CIFAR-10’s native  $32 \times 32$  resolution. Although this input pipeline mirrors how ImageNet-scale ViT architectures are commonly adapted to small images, our model is trained from scratch (no ImageNet pretraining), so the upsampling is an architectural-adaptation choice rather than a requirement of pretrained weights. Upsampling could in principle confound input-gradient statistics, since each input pixel influences many backbone positions. Our CIFAR-native ViT control (Section IV-C) addresses this directly: the native-resolution model is even less sparse and more aligned than the upsampled one, so the CNN-ViT ordering of gradient characteristics is not an upsampling artifact. The control model differs in capacity and patch size, however, so it calibrates the preprocessing confound rather than providing a second matched comparison; the semantic drift analysis likewise reflects the upsampled pipeline.

Our evaluation focuses exclusively on CIFAR-10, and findings may differ on ImageNet where ViTs typically excel and pretrained robust models are available. We evaluate only  $L_\infty$  attacks;  $L_2$  or semantic attacks may reveal different vulnerability patterns. We validate evaluation stability across three random seeds for attack initialization (standard deviation below 0.25 percentage points) and report multiple training runs per architecture (Section IV-E), though additional training runs under the final protocol would be needed to attach uncertainty estimates to the architecture comparison itself. Finally, standard adversarial training for ViT produces lower robust accuracy than specialized techniques such as diffusion-based data augmentation, with which ViTs have recently been reported to surpass WideResNets on small-scale datasets [31], [57]; our ViT results represent baseline adversarial training performance rather than state-of-the-art ViT robustness, and our architecture comparison holds specifically under matched standard adversarial training.

### D. Future Directions

Our findings motivate several research directions. Systematically evaluating CNN-ViT hybrid architectures such as ConViT and CoAtNet could reveal whether combining architectural elements yields robustness benefits beyond either paradigm alone. The gradient sparsity difference we identify—CNN gradients modestly more concentrated under scale-invariant measures, ViT gradients somewhat more spatially distributed—suggests that hybrids might achieve more balanced vulnerability profiles.

Developing training losses that explicitly encourage attention pattern stability under perturbation may address the vulnerability correlates we characterize in ViTs. The early-accumulating semantic drift we observe (largely established by mid-network) suggests that intermediate-layer supervision or consistency regularization could help maintain feature integrity throughout the network.

Investigating whether the robustness gap persists at larger scales (ViT-Base, ViT-Large) and on higher-resolution datasets would establish the generality of our conclusions. Finally, extending our analysis to certified defense methods, including randomized smoothing and interval bound propagation, would determine whether the architectural differences we identify affect achievable certification radii.

## VI. CONCLUSION

This paper presented a behavioral characterization of how CNN and Vision Transformer architectures differ under adversarial attack. Moving beyond simple robustness comparisons, we examined these differences through transfer attack analysis, gradient characterization, and layer-wise feature degradation studies.

Our analysis establishes a statistically significant but modest robustness advantage for the CNN under full-test AutoAttack evaluation (35.74% vs 32.94%, McNemar  $p < 10^{-11}$ ); the WideResNet baseline’s much higher robustness further shows that capacity, training data, and training recipe jointly matter beyond architectural choice. A key methodological finding concerns transfer measurement: the large apparent transfer asymmetry in raw rates (40.2% vs 31.9%) is manufactured entirely by the targets’ unequal clean accuracy and vanishes under the conditioned fooling-rate metric (20.95% vs 20.32%). For this model pair, under PGD-10 at  $\epsilon = 8/255$ , cross-architecture transfer is symmetric—quantifying, in a controlled setting, how large a spurious asymmetry the widely used raw metric can produce when targets differ in clean accuracy. Gradient analysis with scale-invariant measures shows CNN sensitivity is modestly more localized—a difference that survives a native-resolution ViT control—while cross-sample gradient alignment is low for both architectures but moderately higher for the ViT. We also identify a characteristic degradation profile whereby adversarial perturbations cause directional deviation in ViT feature representations that accumulates by mid-network (cosine similarity falling from 0.994 to 0.955) and then plateaus, despite feature-norm changes below 1.5%.

This work characterizes gradient structure alongside cross-architecture transfer behavior, dissociating which gradient properties are architectural (cross-sample alignment) and which are shaped by adversarial training (magnitude and sparsity ordering), and offering methodological guidance in addition to performance comparison. Our systematic layer-wise analysis characterizes how adversarial perturbations propagate through ViT layers, and we distill practical guidance for evaluating and deploying mixed-architecture ensembles. We will release our analysis pipeline to enable future work extending these findings to other architectures and datasets.

Understanding *why* architectures behave differently under adversarial attack—not just *how much*—is essential for devel-

oping targeted defenses. Our analysis shows that the behavioral differences between CNNs and ViTs correlate with scale-invariant differences in gradient structure, particularly how sensitivity is spatially distributed—an effect we verified is not an artifact of gradient scale or input preprocessing—though disentangling architectural effects from capacity and training-recipe factors requires the capacity-controlled follow-ups we outline. Cross-sample gradient alignment is low for both adversarially trained models yet remains moderately higher for the ViT (0.052 vs 0.038, 1.36 $\times$ ); measuring the clean-trained counterparts shows nearly identical levels (0.050 vs 0.030), identifying the alignment gap as an architectural property largely independent of the training regime—while gradient sparsity, whose architecture ordering adversarial training reverses, emerges as the training-shaped characteristic. These dissociations open new questions about the interaction between defense methods and architectural inductive biases. We hope this behavioral perspective guides future research toward more principled defense design.

#### DATA AND CODE AVAILABILITY

The experiments in this study use the CIFAR-10 dataset [46], which is publicly available at <https://www.cs.toronto.edu/~kriz/cifar.html>. Pre-trained WideResNet-28-10 weights were obtained from the RobustBench benchmark [17], [48]. The source code for the analysis pipeline and trained model checkpoints will be made publicly available upon acceptance.

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